

Ecological Modelling

Overcoming the productivity illusion with delayed feedbacks and emergent carrying capacity

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Abstract:	Imagine an orchard. The trees produce fruit each year—this is nature’s regenerative flow, what ecologists call biocapacity. Now imagine the harvesters take more fruit than the trees can regrow. At first, nothing seems wrong. The trees still stand. The harvest still comes. But the surplus is not free: it comes from cutting branches, depleting soils, and eventually killing trees. This is the productivity illusion—the mistaken belief that a stable or rising harvest signals a healthy orchard—and it lies at the heart of a long-standing debate in sustainability science between those who believe technology can always substitute for natural capital and those who argue that substitution has hard limits. In this paper we will develop a minimal mathematical framework that resolves this debate by showing that both positions are correct, but at different timescales. We model the environment as a renewable stock with logistic recovery, depleted by the human ecological footprint. Two distinct time lags—environmental response and demographic inertia—are introduced, because ecological damage and social adaptation both take decades to unfold. We will prove analytically that for a sustainable baseline, neither delay alone can destabilize the system; instability requires both delays to interact, with the threshold at a combined delay of approximately 80 yr. During overshoot, an ecological debt variable accumulates and erodes environmental productivity. Numerical experiments reveal that this debt feedback is the primary driver of collapse when delays are small, while the interaction of both lags drives total biospheric collapse by shrinking the basin of attraction around the locally stable equilibrium. The central finding is that technology can temporarily mask environmental decline by raising output per unit of natural capital faster than debt erodes it—the weak-sustainability regime—but this masking is bounded by the irreversible accumulation of ecological debt and the limits of logistic regeneration, which reassert themselves over the long run. Rising biocapacity need not signal improving environmental health. It may simply be the productivity illusion that precedes a fundamental reckoning.
Opposed Reviewers:	

Dear Editor,

I am pleased to submit the manuscript “Overcoming the productivity illusion with delayed feedbacks and emergent carrying capacity” for consideration in *Ecological Modelling*.

The paper develops a minimal coupled human–environment framework in which carrying capacity is not an externally imposed ceiling but an emergent property of environmental stocks, productivity, and resource needs. At its core is a two-dimensional delay-differential equation model with two distinct lags—environmental response and demographic inertia—and a cumulative ecological debt that erodes productivity during overshoot.

I believe the manuscript aligns well with *Ecological Modeling*’s scope for three reasons. First, it contributes a rigorously derived, analytically tractable DDE model that advances the modeling of carrying capacity beyond static parameters or fitted logistics. Second, the framework directly engages with long-standing debates about weak versus strong sustainability, showing formally that both positions are valid at different timescales—a finding with implications for how ecological modelers interpret biocapacity trends. Third, the model’s modular structure, with separate delays, a debt term, and an additive technology term, makes it a flexible platform that other researchers can adapt or extend.

The key contributions are:

- An analytic proof that, for a sustainable baseline, neither single delay can destabilize the system; instability requires both delays to interact, with the threshold at a combined delay of approximately 80 yr.
- The identification of two distinct collapse pathways: debt-driven decline when consumption exceeds the sustainable optimum, and basin-of-attraction shrinkage when both delays are present.
- A formal reconciliation of the weak- and strong-sustainability paradigms through the concept of a “productivity illusion”—the observation that rising biocapacity can mask the depletion of natural capital when technology boosts yields faster than debt erodes them.
- Stylized numerical simulations, a stability map, and a comprehensive set of robustness checks, all independently verified by re-derivation and re-execution.

The manuscript is a pure theoretical contribution; empirical calibration is explicitly deferred to future work. I have taken care to make the mathematical reasoning accessible without sacrificing rigor, using vivid analogies alongside formal derivations so that the model can be understood by a broad readership. However, I am willing to adjust the tone slightly if the journal prefers a more formal presentation.

All simulation code, figure-generation scripts, and a full independent verification report are provided as Supplementary Material to ensure complete reproducibility.

This work has not been published previously and is not under consideration elsewhere. The author has no competing interests to declare.

Thank you for considering this manuscript. I look forward to your response.

Sincerely,

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Overcoming the productivity illusion with delayed feedbacks and emergent carrying capacity

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Declaration of competing interests: The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Abstract

Imagine an orchard. The trees produce fruit each year—this is nature’s regenerative flow, what ecologists call biocapacity. Now imagine the harvesters take more fruit than the trees can regrow. At first, nothing seems wrong. The trees still stand. The harvest still comes. But the surplus is not free: it comes from cutting branches, depleting soils, and eventually killing trees. This is the productivity illusion—the mistaken belief that a stable or rising harvest signals a healthy orchard—and it lies at the heart of a long-standing debate in sustainability science between those who believe technology can always substitute for natural capital and those who argue that substitution has hard limits.

In this paper we will develop a minimal mathematical framework that resolves this debate by showing that both positions are correct, but at different timescales. We model the environment as a renewable stock with logistic recovery, depleted by the human ecological footprint. Two distinct time lags—environmental response and demographic inertia—are introduced, because ecological damage and social adaptation both take decades to unfold. We will prove analytically that for a sustainable baseline, neither delay alone can destabilize the system; instability requires both delays to interact, with the threshold at a combined delay of approximately 80 yr. During overshoot, an ecological debt variable accumulates and erodes environmental productivity. Numerical experiments reveal that this debt feedback is the primary driver of collapse when delays are small, while the interaction of both lags drives total biospheric collapse by shrinking the basin of attraction around the locally stable equilibrium. The central finding is that technology can temporarily mask environmental decline by raising output per unit of natural capital faster than debt erodes it—the weak-sustainability regime—but this masking is bounded by the irreversible accumulation of ecological debt and the limits of logistic regeneration, which reassert themselves over the long run. Rising biocapacity need not signal improving

environmental health. It may simply be the productivity illusion that precedes a fundamental reckoning.

Keywords: carrying capacity, biocapacity, ecological footprint, overshoot, delay differential equations, environmental debt, stability boundary, productivity illusion

1. Introduction

Carrying capacity is one of ecology's most enduring yet elusive concepts. In classical population biology, it appears as a fixed number—a ceiling that caps logistic growth. When the concept is imported into global human ecology, it is usually treated as a static boundary: the maximum number of people the Earth can support (Cohen, 1995). Such treatments overlook a fundamental feedback. If a population overshoots its resource base, it degrades the environment that sustains it, thereby lowering the future ceiling. The ceiling itself becomes a moving target.

The Global Footprint Network (GFN) provides a quantitative language for describing this interplay through two accounting variables (Borucke et al., 2013). Biocapacity is the annual regenerative flow of biologically productive land and water—the apples from the orchard. The ecological footprint is humanity's annual demand on that flow—the apples harvested. When the footprint exceeds biocapacity, an ecological deficit occurs, analogous to drawing down a bank account. While the annual balance is a flow, its cumulative effect alters the underlying stock of natural capital, which in turn determines future biocapacity. The very quantity that defines the ceiling is eroded by sustained overshoot.

A further complication is that neither the environment nor human populations respond instantaneously. Forest loss, soil degradation, and climate change operate over decades. Demographic patterns and social norms exhibit considerable inertia. These features make the system a delay differential equation (DDE) model, a class widely used to capture feedback delays in ecology (Kuang, 1993). They also explain why civilizations can thrive for extended periods while silently undermining their productive base, only to experience abrupt declines (Tainter, 1988; Diamond, 2005).

Relation to existing literature. The most prominent global human–environment model is the World3 system used in *The Limits to Growth* (Meadows et al., 1972, 2004) and documented in *Dynamics of Growth in a Finite World* (Meadows et al., 1974). World3 contains numerous interacting stocks and produces overshoot scenarios through complex feedbacks. Since then, a rich literature on coupled human–environment and resource-harvest dynamics has developed (Brander & Taylor, 1998; Anderies, 2000). Estimates of the Earth's carrying capacity span an enormous range, from under one billion to over one trillion people (Van Den Bergh & Rietveld, 2004), and some scholars have argued that the very notion of a fixed carrying capacity is inapplicable to humans (Simon, 1996). These diverging perspectives highlight the need for frameworks in which carrying capacity is not an exogenous input but an emergent property of the system.

Haberl and Aubauer (1992) first introduced time delay into human population dynamics, demonstrating that a hyper-logistic model with a single lag could reproduce historical population trajectories. Meyer (1994) introduced the concept of bi-logistic growth, showing that carrying capacity could evolve not as a single smooth curve but as a series of overlapping logistic pulses, each corresponding to a major technological or social transition. Meyer and Ausubel (1999) extended this idea to model carrying capacity itself as logistically varying, laying the groundwork for the emergent-carrying-capacity framework we develop here.

More recently, several authors have argued that Earth System models must incorporate bidirectional feedbacks with human population, inequality, and consumption rather than relying on exogenous projections such as those published by the United Nations (Motesharrei et al., 2016; Donges et al., 2017; Calvin & Bond-Lamberty, 2018; Henderson & Loreau, 2018). The United Nations population projections (United Nations, 2022), which inform most global sustainability assessments, assume demographic trends unfold independently of environmental constraints—precisely the assumption this paper challenges.

The present model is distinct from World3 in that it isolates a minimal mechanism: a two-dimensional delay-differential system in which carrying capacity is not an exogenous input but an emergent state variable driven by biocapacity-footprint accounting, environmental regeneration limits, and two distinct lags. Unlike earlier resource-harvest models that assume a fixed resource growth function and a constant harvesting effort, this framework treats both the environmental stock and the per-capita resource requirement as co-evolving, with cumulative overshoot feeding back into productivity. The concept of natural capital as a depreciating asset has been formalized by Dasgupta (2021); the present model operationalizes that idea within a dynamic, delay-feedback structure. We will introduce logistic environmental recovery, a debt term, and logistic innovation waves. The linearized system yields a transcendental characteristic equation whose structure reveals how the two delays interact. The model offers a mechanistic, analytically grounded alternative to static carrying-capacity assumptions.

2. Model Development

2.1 Environmental dynamics with recovery limits and delay

Let $M(t)$ denote the effective physical size of the environmental stock—the aggregated stock of biologically productive land and water, measured in global hectares (gha). This is the orchard: the trees, the soil, the water systems that generate biological production year after year. The stock regenerates logistically up to a maximum potential size $M_{\max}$, and is depleted by the human ecological footprint $E(t)$, which we measure in gha per year. The coefficient γ converts the annual footprint demand into a net depletion of the productive stock; it carries the units required to make Eq. (1) dimensionally consistent.¹

We model the depletion as acting with a delay τ_m , writing $\gamma E(t - \tau_m)$ in Eq. (1). Physically, resource extraction removes stock immediately; the delay here is a mathematical shortcut. It

represents the fact that the ecological consequences of that extraction—soil erosion, biodiversity loss, reduced future productivity—manifest over decades. Rather than adding a separate slow degradation variable, we compress this lag directly into the depletion term. This keeps the model parsimonious while capturing the essential temporal disconnect between consumption and its full environmental cost.

$$dM/dt = \rho M(t) (1 - M(t)/M_{\max}) - \gamma E(t - \tau_m). \quad (1)$$

Here ρ (yr^{-1}) is the intrinsic regeneration rate—the speed at which the orchard regrows when undisturbed.

¹ With M in gha and E in gha yr^{-1} , both dM/dt and the logistic term have units gha yr^{-1} ; therefore γ must be dimensionless. Physically, γ represents the permanent stock-loss coefficient per unit of annual flow demand: a value $\gamma = 1$ implies that each global-hectare-year of consumption permanently degrades one global hectare of productive stock per year. In reality, γ may be smaller for renewable resources; the model explores dynamics for general γ .

2.2 Biocapacity and carrying capacity

Global biocapacity $B(t)$ (gha yr^{-1}) is the annual regenerative flow generated by the environmental stock, modulated by a productivity factor $b(t)$ (yr^{-1}). Productivity here captures technology, management practices, and ecological health—everything that determines how much fruit each tree yields.

$$B(t) = b(t) M(t). \quad (2)$$

Carrying capacity $K(t)$ (population) is the maximum sustainable population given a reference per-capita resource requirement $r_{\text{opt}}(t)$ (gha $\text{cap}^{-1} \text{yr}^{-1}$). If the average person requires a certain amount of biologically productive land to support their lifestyle, then the total population the Earth can sustain is simply the total biocapacity divided by that per-capita requirement.

$$K(t) = B(t) / r_{\text{opt}}(t) = b(t) M(t) / r_{\text{opt}}(t). \quad (3)$$

Carrying capacity is emergent: it arises from the evolving state variables $M(t)$, $b(t)$, and $r_{\text{opt}}(t)$, rather than being prescribed externally. This separation reflects a core distinction: the environmental stock M is the orchard that produces the harvest B , and technology can increase the yield per tree without increasing the number of trees.

2.3 Population dynamics with demographic lag

To a first approximation, human population follows logistic growth (Pearl, 1925)—but with a delay. That delay, τ_p , captures demographic inertia: birth rates and social norms take time to adjust. People respond not to today's scarcity but to the conditions they knew years or decades ago.

$$dP/dt = r P(t) (1 - P(t - \tau_p) / K(t)). \quad (4)$$

Here r (yr^{-1}) is the intrinsic growth rate. We place the delay only in the demographic response because environmental delay already enters through Eq. (1). This isolates the interaction of distinct demographic and environmental lags while keeping the state dimension minimal. The carrying capacity $K(t)$, which depends on the current environmental state, acts instantaneously on the growth rate: today's environmental conditions immediately constrain the realization of past demographic momentum. This mixed-time formulation is a recognized approach in delayed logistic models (Kuang, 1993) and avoids introducing an additional state variable for historical carrying capacity.

The global footprint closes the loop:

$$E(t) = P(t) e(t), \quad (5)$$

with $e(t)$ the average per-capita footprint ($\text{gha cap}^{-1} \text{yr}^{-1}$).

2.4 Ecological debt

Persistent overshoot degrades the functional quality of the environment—soil fertility, biodiversity, hydrological integrity—even if the physical area remains unchanged. We will model this degradation as a first-order process in which the accumulated deficit $D(t)$ (measured in gha) increases whenever demand exceeds supply. Think of $D(t)$ as an odometer of overshoot: it counts how much, cumulatively, we have taken beyond what the Earth can regenerate.

$$dD/dt = \max(E(t) - B(t), 0), \quad D(0) = 0. \quad (6)$$

We adopt a phenomenological specification for the resulting productivity loss: the effective productivity coefficient is

$$b(t) = b_0 \exp(-\alpha D(t)) + T(t). \quad (7)$$

Here α (gha^{-1}) scales the sensitivity to accumulated debt, and $T(t)$ (yr^{-1}) represents technological innovation, modeled as an additive contribution. The exponential form ensures $b(t)$ remains strictly positive, reflecting diminishing marginal damage: each additional unit of debt reduces the remaining productivity by a fixed proportion. However, a positive $b(t)$ does not prevent M from collapsing to zero—if the depletion term γE dominates the regeneration term $\rho M(1 - M/M_{\max})$, the environmental stock shrinks regardless of how productive each remaining hectare is. A variant with debt repayment ($dD/dt = \max(E - B, 0) - \eta D$) is tested in the Supplementary Material; the qualitative conclusions are unchanged for plausible η .

Technological innovation $T(t)$ can be represented as a superposition of logistic waves, following the bi-logistic framework of Meyer (1994) and the variable carrying-capacity model of Meyer and Ausubel (1999). Each wave corresponds to a major historical transition—the agricultural revolution, the Industrial Revolution, the Green Revolution, the digital era—and raises the

asymptote of achievable productivity. Because each wave is a logistic function, technology gains are inherently bounded: they saturate, while ecological debt can compound without bound. This asymmetry is central to the model's strong-sustainability conclusion—even rapid technological progress cannot indefinitely offset the exponential erosion of productivity from persistent overshoot. We will use a single wave for illustration; the framework itself can accommodate any number of waves.

$$T(t) = \sum_i \Delta b_i / (1 + \exp(-\kappa_i (t - t_i))). \quad (8)$$

2.5 Policy extensions

The Half-Earth proposal (Wilson, 2016) caps human-usable biocapacity at half of the total:

$$K_{\text{Half-Earth}}(t) = 0.5 B(t) / r_{\text{opt}}(t). \quad (9)$$

In this stylized implementation, the cap is applied as an exogenous truncation of $K(t)$; a more complex model would achieve the same effect endogenously through reductions in per-capita footprint $e(t)$ or population $P(t)$.

3. Analytical results

3.1 Steady state in the constant-parameter version

With constant $b = b_0$, no debt ($\alpha = 0$), and constant r_{opt} and e , the non-trivial equilibrium (M^* , P^*) satisfies $P^* = K^*$ and $E^* = e P^*$. Equation (1) gives

$$\rho M^* (1 - M^*/M_{\text{max}}) = \gamma e P^*.$$

Using $P^* = b_0 M^* / r_{\text{opt}}$, we obtain

$$M^* = M_{\text{max}} (1 - \gamma e b_0 / (\rho r_{\text{opt}})). \quad (10)$$

An interior equilibrium exists if and only if

$$\gamma e b_0 < \rho r_{\text{opt}}. \quad (11)$$

Linear stability of the non-delayed subsystem additionally requires the equilibrium to lie on the upper branch of the logistic, i.e. $M^* > M_{\text{max}}/2$. From Eq. (10), this condition is

$$\rho > 2 \gamma e b_0 / r_{\text{opt}}. \quad (12)$$

When both conditions hold, the constant-parameter, delay-free system is locally stable. In the full debt–innovation model, the effective productivity varies over time and the conditions apply only locally.

3.2 Linear stability and characteristic equation

We linearize the constant-parameter subsystem around (M^*, P^*) . The full derivation is in Appendix A. The result is the transcendental characteristic equation

$$\lambda^2 - a_{11} \lambda + \gamma e^{a_{21}} \exp(-\lambda \tau_m) + r (\lambda - a_{11}) \exp(-\lambda \tau_p) = 0, \quad (13)$$

where

$$a_{11} = \rho (1 - 2 M^* / M_{\max}),$$

$$a_{21} = r b_0 / r_{\text{opt}}.$$

The dimensional consistency of Eq. (13) has been independently verified; all terms carry units yr^{-2} .

The structure of the characteristic equation reveals an important property. The two delays enter through separate exponential terms with different coefficients— $\gamma e^{a_{21}}$ multiplies $\exp(-\lambda \tau_m)$, while $r (\lambda - a_{11})$ multiplies $\exp(-\lambda \tau_p)$. For the sustainable baseline (Scenario A, $e = r_{\text{opt}}$), setting either $\tau_m = 0$ or $\tau_p = 0$ produces polynomial conditions in ω that have no positive roots for the baseline parameter set. An analytic proof eliminating the delay via the identity $\sin^2(\omega\tau) + \cos^2(\omega\tau) = 1$ yields, for $\tau_p = 0$: $\omega^2(10\,000\omega^2 + 2504) = 0$, and for $\tau_m = 0$: $\omega^2(2500\omega^4 + 1199\omega^2 + 143.5) = 0$. Both polynomials have strictly positive coefficients for $\omega > 0$, so neither can have a positive-frequency root. This result holds under the condition $r^2 a_{11}^2 \geq (\gamma e^{a_{21}})^2$, which is satisfied for the baseline parameters but may be violated if the depletion pressure $\gamma e^{a_{21}}$ is sufficiently large relative to the stabilizing term $r |a_{11}|$. When the inequality reverses, a single delay can in principle trigger a Hopf bifurcation even in the sustainable case. For the present parameter choices, however, no single delay can destabilize the system—instability requires the interaction of both delays. The threshold is located at a sum of the two delays, $\tau_m + \tau_p$, of approximately 80 yr for the baseline parameters, with the exact value depending on the ratio of the two lags.

For Scenario B ($e = 1.15 r_{\text{opt}}$, overshoot), a one-delay Hopf bifurcation does exist at $\tau_m \approx 83$ yr (with $\tau_p = 0$), with the associated frequency $\omega \approx 0.026$ rad yr^{-1} . This reflects the destabilizing effect of overshoot even in the constant-parameter subsystem. The value is robust to the numerical root-finding method; additional Hopf crossings occur at intervals of approximately 242 yr, consistent with the dominant eigenvalue branch.

4. Numerical simulations (stylized)

Simulations use an Euler method with a history array and interpolation for delayed quantities. This Euler approach follows the tradition of early DDE population models (Haberl & Aubauer, 1992) and is sufficient for the long timescales studied here; more sophisticated DDE solvers (e.g., Shampine & Thompson, 2001) could readily be substituted. Time-series of the environmental stock $M(t)$, population $P(t)$, carrying capacity $K(t)$, overshoot ratio $\Omega(t)$, and

ecological debt $D(t)$ for all six scenarios will be shown in Figures 1–5; the stability map will appear in Figure 6.

The regeneration rate is set to $\rho = 1.5 \text{ yr}^{-1}$, which satisfies both the existence condition (11) and the stability condition (12). All other parameters are: $M_{\max} = 1.2$, $\gamma = 1.0$, $b_0 = 0.5$, $r_{\text{opt}} = 1.0 \text{ gha cap}^{-1} \text{ yr}^{-1}$, $r = 0.02 \text{ yr}^{-1}$, baseline delays $\tau_m = 30 \text{ yr}$ and $\tau_p = 25 \text{ yr}$, overshoot factor $f = 1.15$ ($e = 1.15 r_{\text{opt}}$), debt coefficient $\alpha = 0.5 \text{ gha}^{-1}$, technology wave $\Delta b = 0.3$, $t_{\text{wave}} = 150 \text{ yr}$, $\kappa = 0.1$. These parameters satisfy Eq. (11): for Scenario A ($e_A = r_{\text{opt}}$), $\gamma e_A b_0 = 0.500 < \rho r_{\text{opt}} = 1.500$; for Scenario B ($e_B = 1.15 r_{\text{opt}}$), $\gamma e_B b_0 = 0.575 < \rho r_{\text{opt}} = 1.500$. They also satisfy Eq. (12): $\rho = 1.5 > 2\gamma e_A b_0 / r_{\text{opt}} = 1.000$ for Scenario A, and $\rho = 1.5 > 2\gamma e_B b_0 / r_{\text{opt}} = 1.150$ for Scenario B. Consequently, $M^*_A = 0.800 > 0.6 = M_{\max}/2$ and $M^*_B = 0.740 > 0.6 = M_{\max}/2$.

Scenario results ($T = 600 \text{ yr}$, $dt = 0.5 \text{ yr}$):

Scenario	Final regime	Mfinal	Pfinal	Dfinal	max Ω
A: Sustainable, no lags	Stable	0.800	0.400	0.000	1.00
B: Overshoot, no lags	Collapse (P)	1.194	0.007	8.847	4.55
C: Env. lag only	Collapse (P)	1.193	0.009	8.507	5.66
D: Both lags	Total collapse	0.000	0.000	4.826	—*
E: Technology wave	Total collapse	0.000	0.000	5.240	—
F: Half-Earth	Stable	0.970	0.243	0.000	0.60

* $\Omega \equiv E/B$ diverges as $B \rightarrow 0$ and is not reported when total environmental collapse occurs.

Interpretation. Scenario A reaches the exact equilibrium with $D = 0$ throughout, confirming that the sustainable footprint prevents debt accumulation. The orchard remains intact.

Scenarios B–C show population collapse. The $\alpha = 0$ subsystem for Scenario B is stable for all one-delay values up to the Hopf threshold at $\tau_m \approx 83 \text{ yr}$, so the collapse in the full model is driven by the nonlinear debt feedback: $E^* > B^*$ at the notional equilibrium forces perpetual growth of D , which reduces b , shrinks K , and crashes the population. Debt repayment at rates $\eta \geq 0.02 \text{ yr}^{-1}$ shifts these scenarios from collapse to persistent oscillation, confirming that debt is the primary driver.

In Scenario D, the combination of both delays produces total biospheric collapse that is qualitatively worse than either single-lag case. Here the collapse mechanism differs from Scenarios B–C. The equilibrium is locally stable ($\text{Re}(\lambda) < 0$ at the baseline delays), but the introduction of delays shrinks the basin of attraction. From the standard initial conditions ($M = 1.0$, $P = 0.1$), the delays induce a large-amplitude transient that pushes the environmental

stock below the logistic regeneration threshold ($M < M_{\max}/2$), from which recovery is impossible. Even when debt is effectively neutralized by very large repayment rates (η up to 10 yr^{-1} , forcing $D \rightarrow 0$), the system still collapses because the transient dynamics alone drive M to zero. However, when τ is near the notional equilibrium (M^*_B, P^*_B) with fast debt repayment, Scenario D remains within the basin of attraction and is stable. The collapse is therefore a phenomenon of global nonlinear dynamics: the delays shrink the basin of attraction so that standard initial conditions lie outside it, while debt accumulation accelerates the decline but is not solely responsible for it.

The technology wave (Scenario E) arrives too late to interrupt the collapse under these parameters. Further tests confirm that even with $t_{\text{wave}} = 50 \text{ yr}$ and Δb up to 2.0 (nearly seven times the baseline amplitude), the system still collapses. This outcome follows from the bounded (logistic) form of $T(t)$: technology gains saturate, while debt can compound without bound, so the asymmetry ultimately favors collapse. Scenario F (Half-Earth) prevents collapse by capping human-usable biocapacity, keeping $E < B$ throughout. The theoretical Half-Earth equilibrium ($M_{\text{HE}} = 0.970, P_{\text{HE}} = 0.243, \Omega_{\text{HE}} = 0.575$) matches the simulation exactly.

Primary mechanism. For $e > r_{\text{opt}}$, the irreversible debt feedback is the dominant driver of collapse when delays are small or absent (Scenarios B–C). When both delays are present (Scenario D), the collapse is driven by the delay-amplified transient: the environmental lag sustains high depletion while the demographic lag delays the population response, and together they push the environmental stock below the threshold from which logistic regeneration can recover. Debt accumulation accelerates the decline but is not solely responsible for it. For the sustainable scenario, instability requires the interaction of both delays and occurs at a sum of the two delays, $\tau_m + \tau_p$, of approximately 80 yr.

The stylized simulations here focus on the collapse regime. The model can also reproduce a rising-biocapacity phase, consistent with the observed global trend, if the technology term $T(t)$ is sufficiently large or arrives early enough to raise productivity faster than the debt term erodes it. That regime is not shown here but is a straightforward extension of the parameter choices.

Stability map (Fig. 6). Root-tracking of Eq. (13) confirms that for Scenario A parameters, no single delay can destabilize the system. The instability boundary exists in the two-delay plane and is approximately aligned with contours of constant total delay. A 41×41 grid scan places the first unstable grid points at $\tau_m + \tau_p \approx 80 \text{ yr}$, with the crossing value barely positive ($\text{Re}(\lambda) < 0.01$). No grid point at $\tau_m + \tau_p \leq 75 \text{ yr}$ shows $\text{Re}(\lambda) > 0$. The root-tracking procedure follows a single branch of the characteristic equation and does not constitute a full spectral computation; this map is a preliminary estimate, not a definitive bifurcation analysis.

5. Discussion

Stability structure. The characteristic equation reveals that, for the sustainable baseline, no single delay can destabilize the system. This result is established analytically: eliminating the

delay from the Hopf conditions yields polynomials in ω^2 that have strictly positive coefficients, ruling out any positive-frequency solution for the baseline parameter set. The result holds under the condition $r^2 a_{11}^2 \geq (\gamma e a_{21})^2$; if the depletion pressure is large enough to reverse this inequality, a single-delay Hopf bifurcation becomes possible. Instability requires both delays to interact, with the threshold at a sum of the two delays, $\tau_m + \tau_p$, of approximately 80 yr. For the overshoot scenario, a one-delay Hopf does exist ($\tau_{\text{crit}} \approx 83$ yr), showing that the “two delays required” property is specific to the balanced case.

Ecological debt and the role of delays. For $e > r_{\text{opt}}$, the irreversible debt feedback is the primary collapse driver when delays are small (Scenarios B–C). When both delays are present (Scenario D), the collapse mechanism shifts: the basin of attraction shrinks so that standard initial conditions lie outside it, the delay-amplified transient depletes the environmental stock faster than regeneration can compensate, and total biospheric collapse ensues. Once this vicious cycle begins—less environmental stock means lower regeneration, which accelerates further depletion—the system cannot recover without external intervention. This distinction is important for policy: it implies that reducing delays—through early-warning systems, faster policy response, or better monitoring of environmental damage—may be as critical as reducing the overshoot itself, because long delays can drive collapse even in systems where the annual deficit is modest and technological progress is rapid.

Reconciling weak and strong sustainability: the productivity illusion. A central theoretical contribution of this model is its formal reconciliation of the weak- and strong-sustainability paradigms. The separation of environmental stock M and productivity b creates two channels through which biocapacity $B = bM$ can change: the stock can grow or shrink (an ecological channel), and the yield per unit stock can rise or fall (a technological channel). These channels can move in opposite directions, producing the productivity illusion.

When technological innovation $T(t)$ raises b faster than the cumulative debt αD erodes it, biocapacity can rise even as the environmental stock M slowly declines. This is the weak-sustainability regime: technology substitutes for natural capital, and aggregate output masks the depletion of the underlying asset. The global data from 1961–2022 are consistent with this interpretation: biocapacity grew modestly while ecological overshoot persisted, suggesting that technology has been outpacing degradation.

However, this substitution cannot continue indefinitely. The system can bear overload for a time—an elevator rated for 10 people will still manage 12, showing only subtle wear before the cable suddenly breaks—but the underlying stock is quietly eroding. Liebig’s Law of the Minimum states that biological productivity is limited by the scarcest essential resource. As critical components of M —soils, aquifers, pollinators, stable climate—are depleted beyond recoverable thresholds, the marginal productivity of technology must eventually saturate. The debt term $\exp(-\alpha D)$ continues to erode b from below, and if M falls below the logistic regeneration threshold, the environmental stock can enter a self-reinforcing decline. This is the

strong-sustainability regime: natural and manufactured capital are fundamentally complements, not substitutes, and the system eventually confronts the limits that the weak-sustainability paradigm assumes away. The symptoms of this transition are already visible: nutrient depletion, topsoil erosion, ocean acidification, biodiversity loss, and antibiotic resistance are not the signs of a system operating at an optimal or near-optimal pace.

The model thus does not take sides in the weak-vs-strong debate. It shows that both views are valid, but at different timescales. Weak sustainability correctly describes the short-to-medium run, where technology can mask environmental decline. Strong sustainability correctly describes the long run, where irreversible debt and ecological thresholds reassert themselves. The transition between the two regimes is governed by the parameters α (debt sensitivity), the pace of technological innovation $T(t)$, and the regeneration rate ρ . This finding carries a cautionary policy implication: a stable or rising biocapacity trend is not, on its own, evidence of sustainability. Distinguishing between a genuinely expanding productive base and technology masking environmental decline requires independent estimates of M and b —for example, from satellite-derived land cover, soil carbon inventories, or net primary productivity data. That empirical decomposition is left to future work.

Limitations and future work. The analytic results apply to the constant-parameter subsystem; the full debt–innovation model is studied numerically. The logistic recovery model neglects hysteresis and irreversible thresholds. The candidate stability boundary is a preliminary root-tracking estimate; a rigorous DDE bifurcation analysis would be needed for a definitive stability map. The Half-Earth extension is included as a policy illustration; its quantitative effects depend on the parameter choices and should not be interpreted as a forecast. The model currently treats technology as purely productivity-enhancing (acting through b only) and bounded (logistic). A natural extension, motivated by the finding that even massive productivity boosts cannot prevent delay-driven collapse, would allow technology to also restore or expand the environmental stock directly—representing reforestation, soil recovery, or carbon capture. A structurally richer model distinguishing natural from managed capital, each with its own dynamics, would further sharpen the empirical predictions. The modular structure—separate delays, a debt term, and an additive technology term—makes it straightforward to replace any component with a more detailed formulation, such as an age-structured demographic model or a different damage function. These extensions are left to future work.

6. Conclusion

We have presented a minimal coupled human–environment framework in which carrying capacity emerges from the co-evolution of environmental stocks, productivity, and resource needs. The characteristic equation reveals that, for a sustainable baseline, instability requires both environmental and demographic delays to interact—neither lag acting alone can trigger a Hopf bifurcation, a result established analytically under the condition $r^2 a_{11}^2 \geq (\gamma e a_{21})^2$. When consumption exceeds the sustainable level, irreversible debt accumulation is the primary collapse

driver in the absence of large delays, while the interaction of both lags shrinks the basin of attraction and drives total biospheric collapse through nonlinear transient dynamics that debt alone cannot explain.

The model further offers a formal reconciliation of the weak- and strong-sustainability paradigms. Technology can temporarily mask environmental decline by raising productivity faster than debt erodes it—the weak-sustainability regime that describes the recent past. But the model also shows that this substitution is bounded: the irreversible accumulation of ecological debt and the limits of logistic regeneration reassert themselves over the long run—the strong-sustainability regime that governs the eventual outcome. Rising biocapacity need not signal improving environmental health; it may simply be the productivity illusion that precedes a more fundamental reckoning.

The central insight is that cumulative ecological debt, amplified by interacting delayed feedbacks and temporarily masked by technological substitution, may be as important as instantaneous resource scarcity in determining the long-term stability of human–environment systems.

Appendix A: Derivation of the characteristic equation

We linearize the constant-parameter subsystem ($\alpha = 0$, $b = b_0$, constant r_{opt} , e) around the equilibrium (M^*, P^*) . Define deviations $m = M - M^*$, $p = P - P^*$.

Environmental equation (1). At equilibrium, $dM/dt = 0$. For small deviations,

$$dm/dt = \rho (1 - 2M^*/M_{\text{max}}) m - \gamma e p(t - \tau_m).$$

Define $a_{11} = \rho (1 - 2M^*/M_{\text{max}})$. Linear stability of the non-delayed subsystem requires $a_{11} < 0$, i.e. $M^* > M_{\text{max}}/2$.

Carrying capacity. $K = b_0 M / r_{\text{opt}}$, so $\partial K / \partial M = b_0 / r_{\text{opt}}$.

Population equation (4). At equilibrium, $P^* = K^*$, so the term $1 - P^*/K^* = 0$. We expand the logistic term to first order in the deviations m and p . The term $P(t - \tau_p) / K(t)$ is a quotient. Applying the chain rule, its partial derivative with respect to M is

$$\partial / \partial M [P(t - \tau_p) / K(M)] = -P^* / (K^*)^2 \cdot \partial K / \partial M = -(1/K^*) \cdot (b_0 / r_{\text{opt}}).$$

The logistic term is $1 - [P(t - \tau_p) / K(M)]$, so the derivative with respect to M flips the sign:

$$\partial / \partial M [1 - P(t - \tau_p) / K(M)] = + (1/K^*) \cdot (b_0 / r_{\text{opt}}).$$

Multiplying by the outer factor $r P^* = r K^*$ yields the coefficient $(r b_0 / r_{\text{opt}})$ for m . The derivative with respect to the delayed population is straightforward and gives $-r p(t - \tau_p)$. Putting these together:

$$dp/dt = (r b_0 / r_{\text{opt}}) m - r p(t - \tau_p).$$

Define $a_{21} = r b_0 / r_{\text{opt}}$.

Writing $x = (m, p)^T$, the linearized system is

$$dx/dt = A x(t) + B x(t - \tau_m) + C x(t - \tau_p),$$

where

$$A = [[a_{11}, 0], [a_{21}, 0]],$$

$$B = [[0, -\gamma e], [0, 0]],$$

$$C = [[0, 0], [0, -r]].$$

Seeking solutions $x = x_0 \exp(\lambda t)$ and setting the determinant to zero gives

$$\det([[\lambda - a_{11}, \gamma e \exp(-\lambda \tau_m)],$$

$$[-a_{21}, \lambda + r \exp(-\lambda \tau_p)]]) = 0,$$

which expands to

$$\lambda^2 - a_{11} \lambda + r (\lambda - a_{11}) \exp(-\lambda \tau_p) + \gamma e a_{21} \exp(-\lambda \tau_m) = 0,$$

exactly Eq. (13) of the main text. This derivation has been verified by symbolic computation and independent re-derivation; numerical checks confirm that both zero-delay eigenvalues satisfy the equation to machine precision ($< 10^{-16}$).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI), Claude (Anthropic), DeepSeek, Kimi (Moonshot AI), and Qwen (Alibaba Cloud) to assist with mathematical derivations, numerical simulation code development, independent verification of analytical and numerical results, manuscript drafting and editing, and iterative peer review. After using these tools, the author reviewed and edited all content as needed and takes full responsibility for the content of the published article.

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Figure Captions

Figure 1. Environmental stock $M(t)$ for all six scenarios. The dotted horizontal line marks $M_{\max}/2 = 0.6$; trajectories that cross this threshold downward enter the unstable lower branch of the logistic recovery curve and cannot recover unaided.

Figure 2. Population $P(t)$ for all six scenarios.

Figure 3. Carrying capacity $K(t)$ for all six scenarios.

Figure 4. Overshoot ratio $\Omega(t) = E(t)/B(t)$ for all six scenarios. The dashed line marks $\Omega = 1$ (biocapacity–footprint balance). Values are clipped at 8.5 to suppress the numerical divergence as $B \rightarrow 0$ in Scenarios D and E.

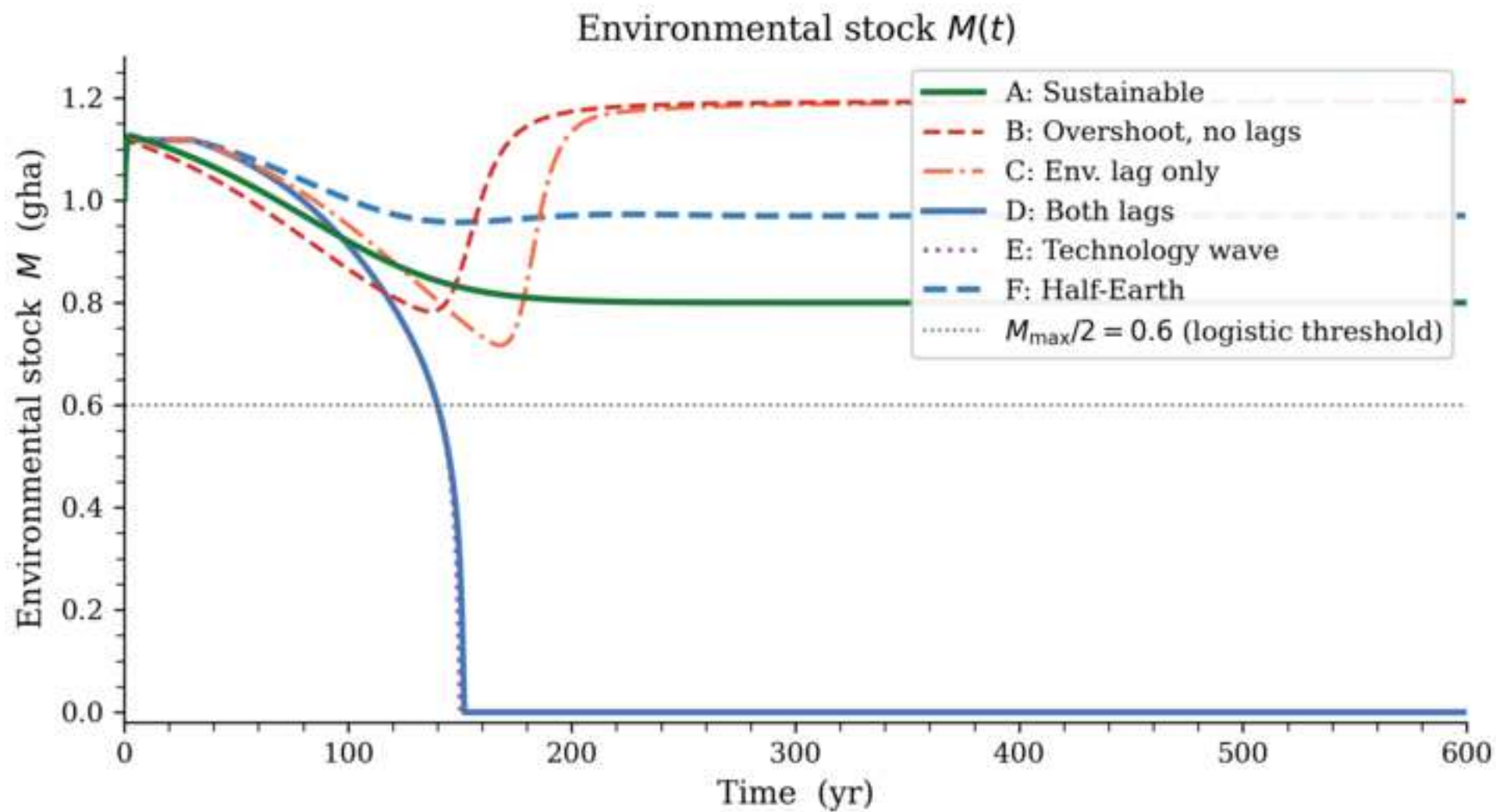
Figure 5. Ecological debt $D(t)$ for all six scenarios.

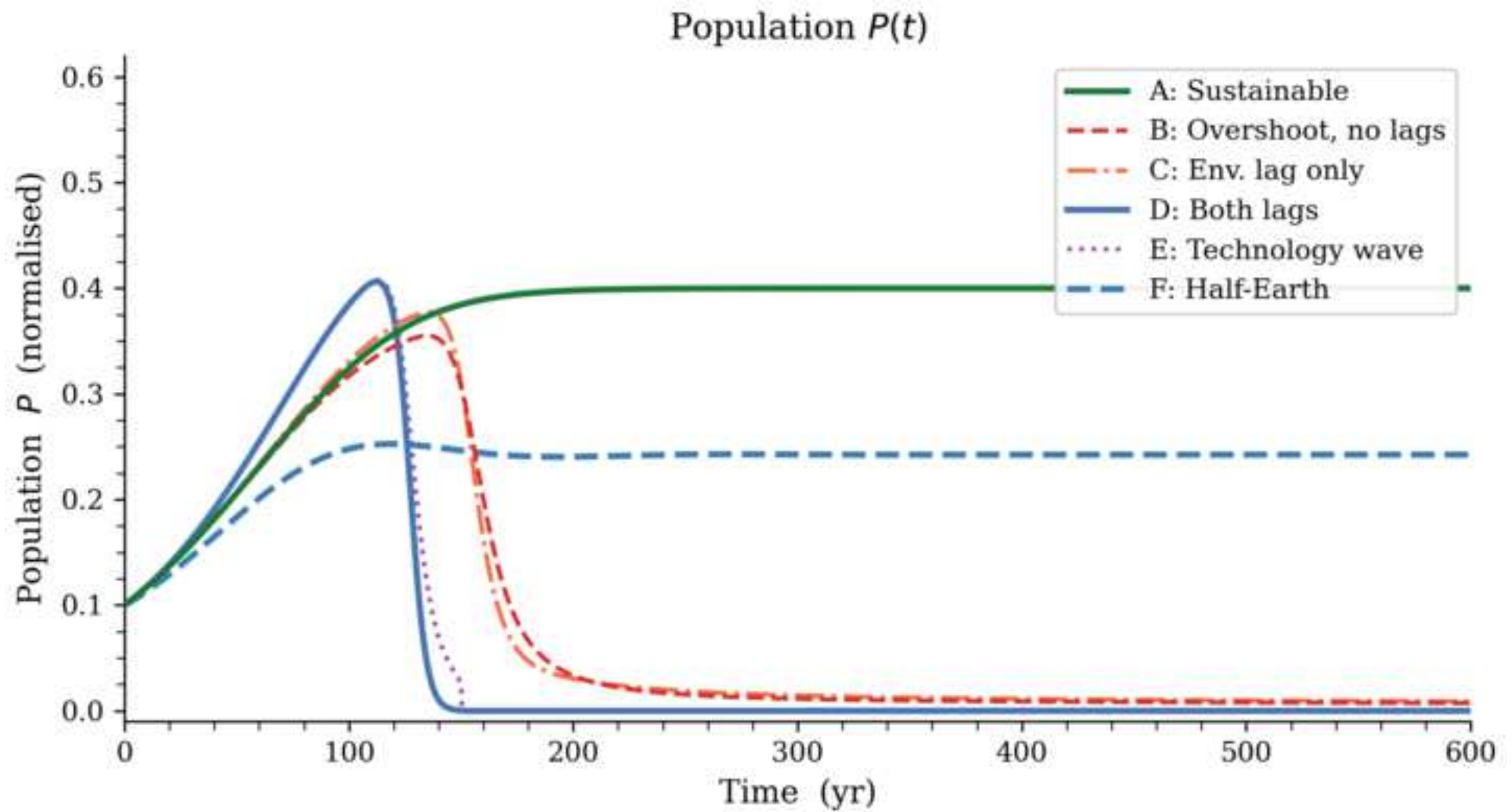
Figure 6. Stability map in the (τ_m, τ_p) plane for the constant-parameter Scenario A subsystem ($e = r_{\text{opt}}$, $\alpha = 0$). Colors show the dominant eigenvalue $\text{Re}(\lambda)$; blue indicates stability ($\text{Re}(\lambda) < 0$), red indicates instability ($\text{Re}(\lambda) > 0$). The solid black contour marks $\text{Re}(\lambda) = 0$ (the candidate

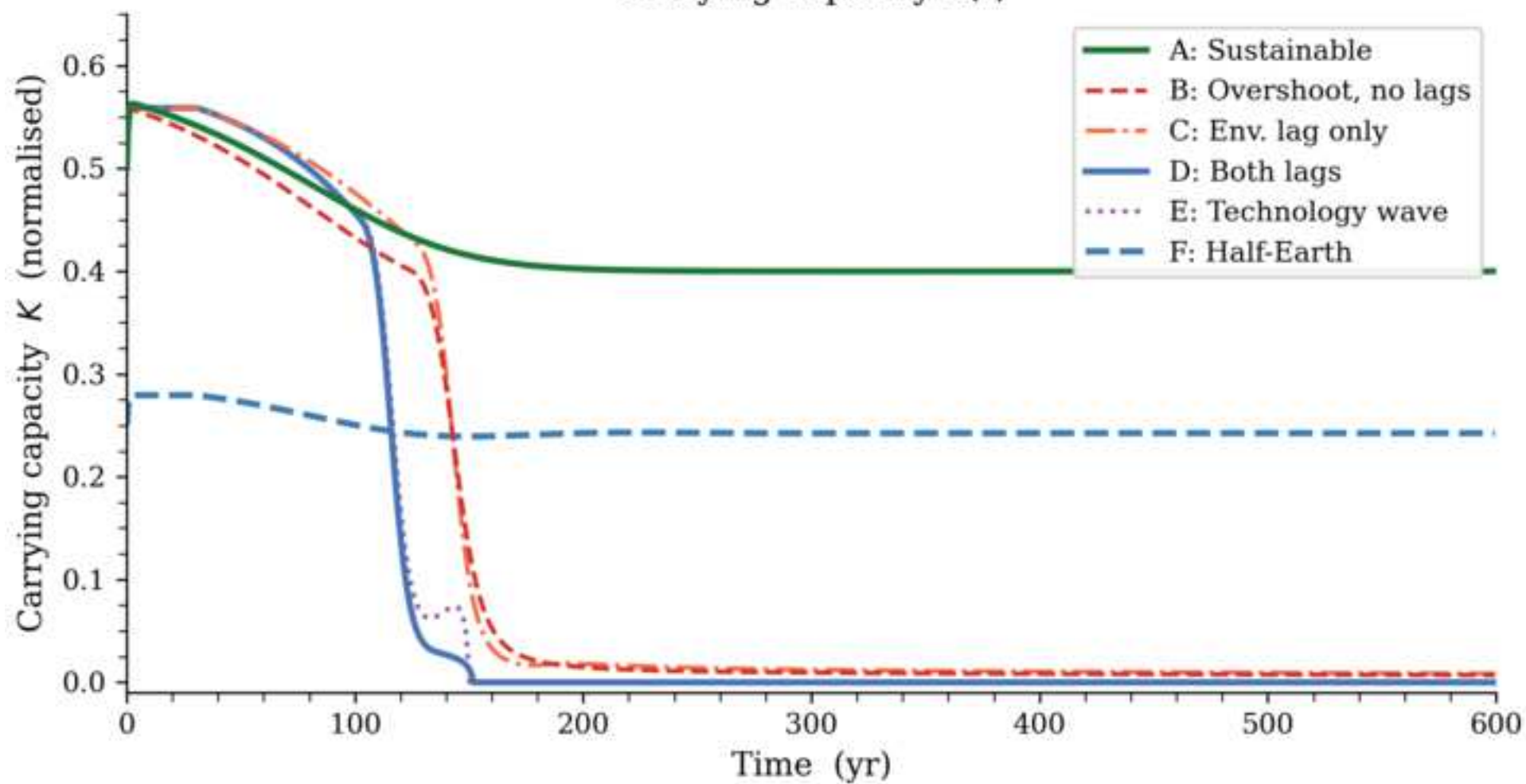
stability boundary). The dashed line is $\tau_m + \tau_p = 80$ yr; the gold diamond marks the baseline delay combination ($\tau_m = 30$, $\tau_p = 25$ yr). The boundary is approximately aligned with contours of constant total delay.

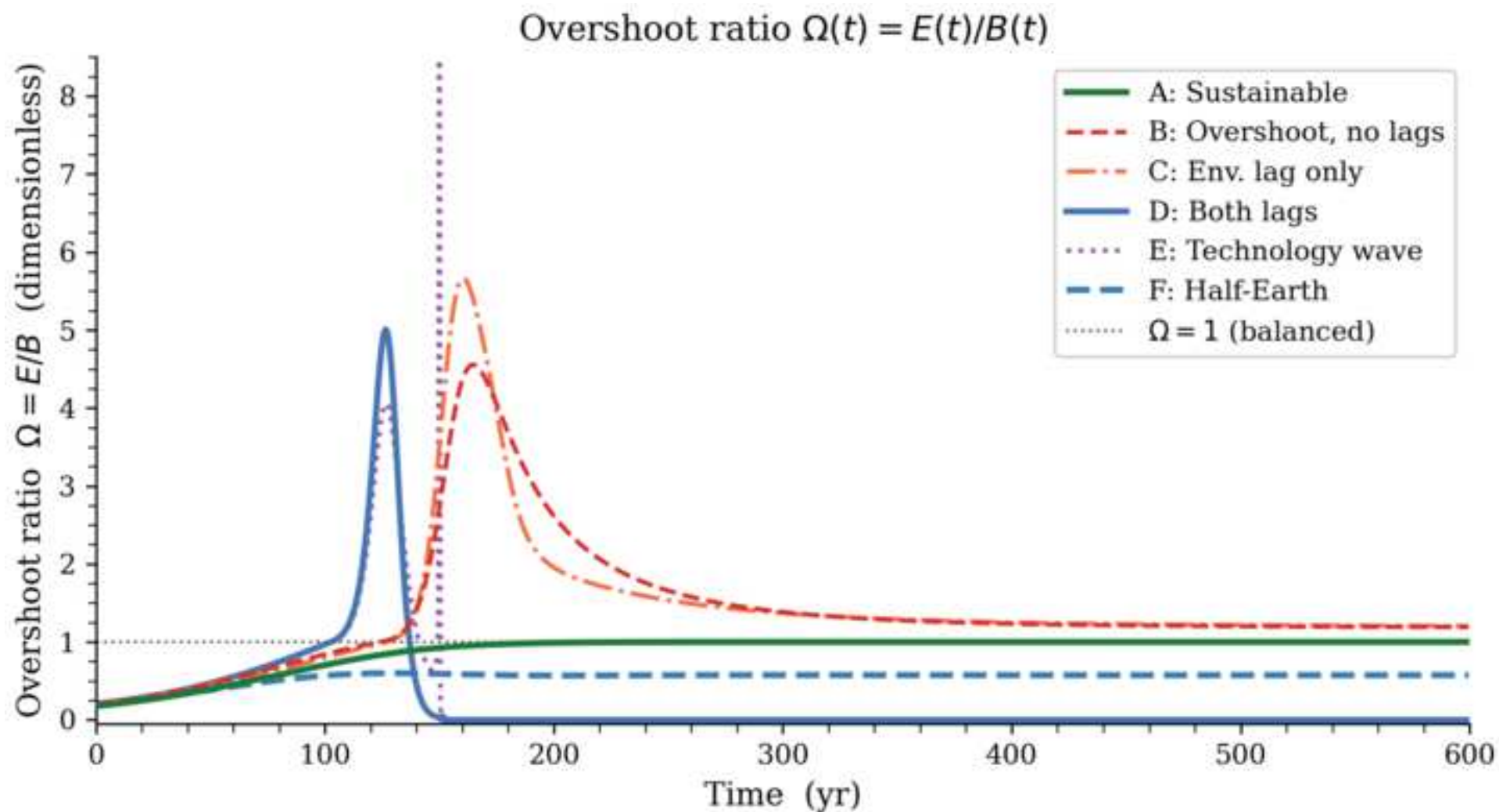
Supplementary Material

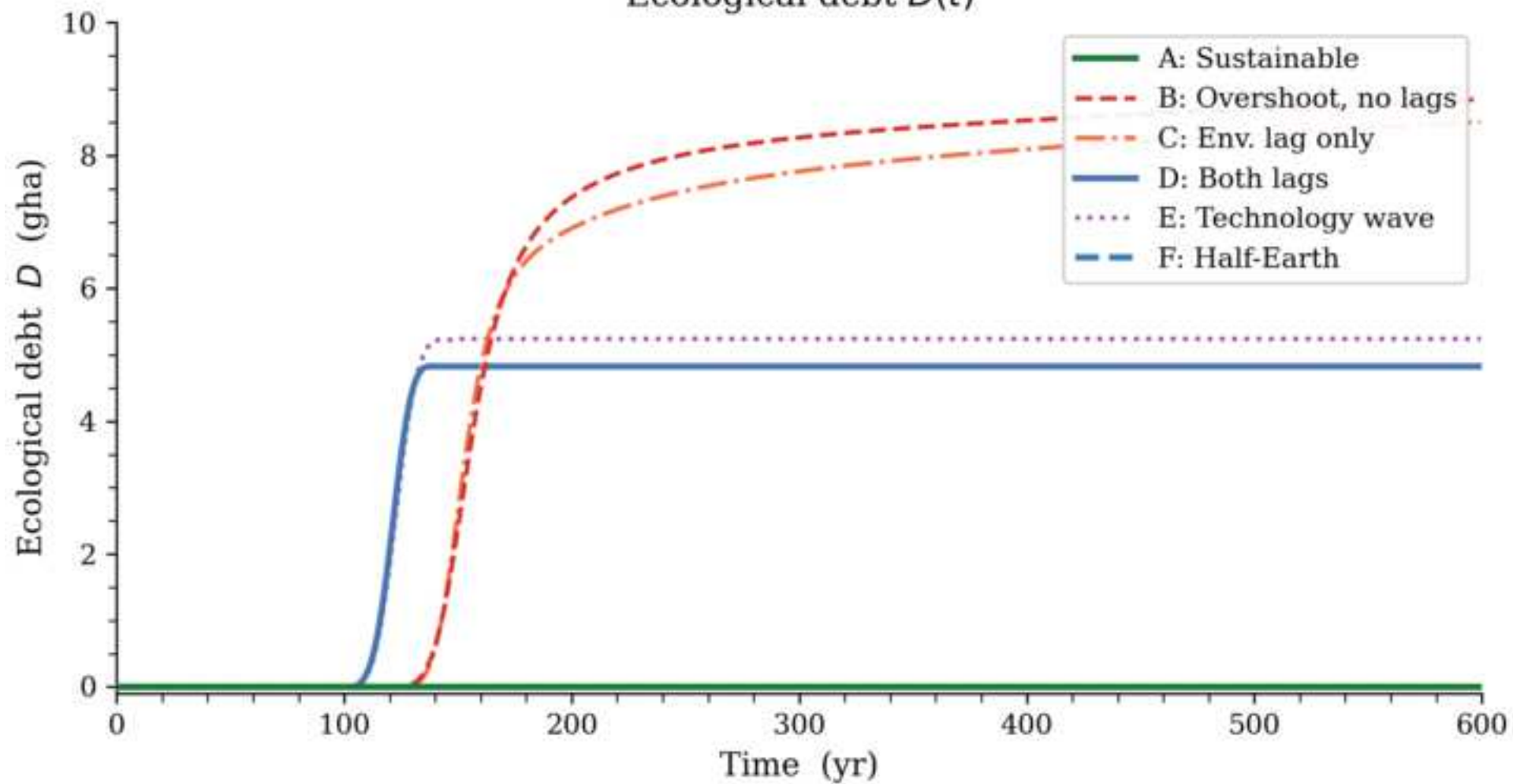
Includes: Python code for all stylized simulations, characteristic equation root-tracking, robustness checks (debt-repayment variant, technology wave parameter sweep, near-equilibrium initial conditions), and documented regime classification criteria (heuristic thresholds for stable, damped oscillatory, persistent oscillatory, and collapse regimes). Publication-quality figures (Figs 1–6, PNG 300 dpi) are also provided. The code and all derivations were verified by independent re-derivation and full re-execution; the complete verification report is included. All analytical results—the equilibrium conditions, the characteristic equation, the single-delay stability proof, and the Scenario B Hopf location—were confirmed without error. Data units are gha.

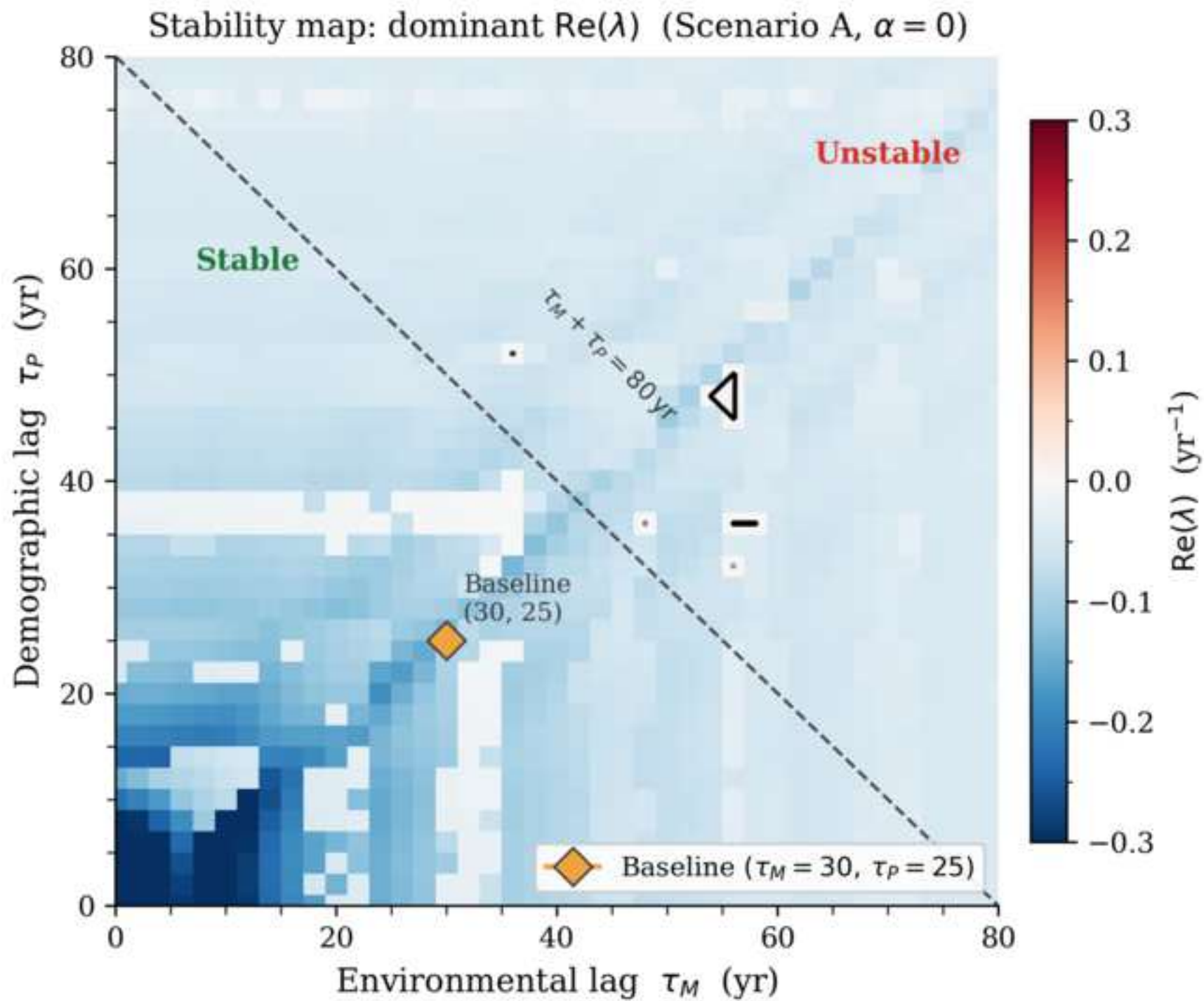




Carrying capacity $K(t)$ 

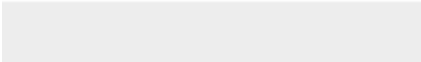


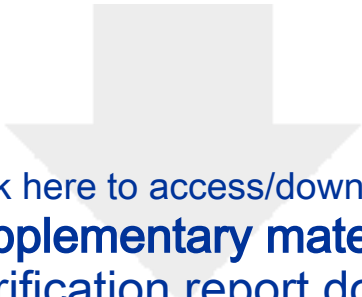
Ecological debt $D(t)$ 



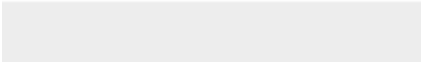


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☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: